

Akbar Aman

Systems & AI Infrastructure Engineering Candidate

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SUMMARY

Computer Engineer (Senior standing, May 2027) focused on Systems and AI Infrastructure, with experience designing constrained system architectures, building production backends, and conducting hypothesis-driven applied research under real-world performance and constraints.

EDUCATION

University of Illinois Chicago, College of Engineering Expected: May 2027
B.S. in Computer Engineering (Computer & Networked Systems Track), Minor in Mathematics GPA: 3.59
– **Relevant Coursework:** Comp Arch, Networking, Data Structs & Algos, Digital Systems, Systems Programming
– **Leadership:** Computer Science Teaching Assistant (MATLAB, 2 terms) & Engineering Success Mentor (2 terms).

EXPERIENCE

Quantitative Research & AI/ML Intern Jan 2026 – Present
Crypt0nest.io (Tekly Studio) Remote
– Engineered ML data pipelines for algorithmic trading research, enforcing strict backtesting and preventing look-ahead bias on time-series market data. Improved experimentation throughput by implementing vectorized featurization.

AI Systems Engineer Oct 2025 – Present
Talent Strategy Experts Remote
– Architected & implemented production version of a multi-tenant SaaS platform for AI inference, owning end-to-end backend, frontend, deployment & system documentation, & preparing the system for team-based development & scale. Designed isolation boundaries, role-based access control, and reliability safeguards to support secure org-level data.

Founding Engineer Sep 2025 – Present
ArkBoosted LLC Remote
– Developing client-run AI infrastructures; Built a scalable multi-tenant RAG system for a SaaS platform serving 700+ organizations, utilizing PostgreSQL (pgvector) & LLMs to optimize vector retrieval latency and query throughput.
– Implemented an automated performance benchmarking engine utilizing algorithmic scoring to audit network latency and resource bottlenecks, driving optimization strategies that improved client operational efficiency by over 90%.

LLM Evaluation Specialist Sep 2025 – Present
Alignerr Remote
– Executed large-scale LLM evaluation and alignment across reasoning, coding, and function-calling domains, applying rubric-based and human-preference assessments on thousands of samples to improve model accuracy and reliability.

Deep Learning/AI Research Intern Jun 2024 – Aug 2024
UIC Department of Electrical & Computer Engineering Chicago, IL
– Led a team of five to design and prototype *Vision Mamba*, a custom neural network achieving a 30% accuracy improvement while optimizing for low-latency, high-throughput inference on constrained edge devices, and completed specialized advanced HW/SW co-design training using hls4ml to synthesize PyTorch models into FPGA bitstreams.

TECHNICAL PROJECTS

Local Quantized LLM Inference Engine | *Systems Architecture* [GitHub] Jan 2026 – Present
– Designed a **CPU-only, offline** quantized LLM inference architecture under a strict $\leq 8\text{GB}$ RAM constraint, analyzing memory hierarchy bottlenecks, KV-cache behavior, and context growth policies to reason about prompt assembly, bounded context windows, and truncation strategies required to maintain predictable latency and resource usage.

Bare-Metal Embedded Safety System | *C, ARM Assembly, CCS, Cortex-M4F* [GitHub]
– Engineered a **dual-mode** bare-metal system on ARM Cortex-M4F utilizing Timer Capture ISRs, ADC sequencers, and Memory Mapped I/O-based PWM drivers to achieve precision ranging and 0.1°F resolution, while implementing a non-blocking architecture via Wait-For-Interrupt (WFI) to minimize power consumption during idle execution cycles.

Full-Stack MLS Property Platform (IDX Exchange Internship) | *Python, PHP, MySQL, REST API* [Live]
– **Led a SWE team of 6;** Created a real-time property search platform integrating live MLS data via Restful APIs utilizing OAuth 2.0; engineered automated cron-job synchronization and optimized MySQL schemas to manage 1,000+ active listings on a Linux VPS environment while ensuring efficient data normalization and low-latency.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, Java, SQL (PostgreSQL/MySQL), PHP, Bash, ARM Assembly
Frameworks & Web: React, FastAPI, Node.js, REST APIs, WebSockets, HTML/CSS, OAuth, Nginx, Terraform
Systems & AI Tools: Linux, Docker, AWS, CI/CD, Git, PyTorch, GGUF, Performance Modeling, FPGA (Vivado)